Week 3: Sequences (Part 2)

F18CD

Monotone Sequences

Definition 7.1. Monotone Sequence

A sequence (a_n) is said to be *monotone increasing* if $a_n \leq a_{n+1} \forall n \in \mathbb{N}$.

Similarly, a sequence (b_n) is said to be *monotone decreasing* if $b_n \geq b_{n+1} \forall n \in \mathbb{N}$.

If a sequence is either monotone increasing or monotone decreasing, then it is said to be *monotone.*

Increasing: $a_{\text{later}} \geq a_{\text{now}}$

Decreasing: $a_{\text{before}} \geq a_{\text{now}}$

Example 7.2. Show that $\left(a_n\right)=\left(\sqrt{n}\right)$ is monotone increasing.

Proof. $a_{n+1}-a_n=\sqrt{n+1}-\sqrt{n}=\left(\sqrt{n+1}-\sqrt{n}\right)\dfrac{\sqrt{n+1}+\sqrt{n}}{\sqrt{n+1}+\sqrt{n}}=\dfrac{1}{\sqrt{n+1}+\sqrt{n}}>0$

$\forall n\in\mathbb{N}$. Hence $a_{n+1}>a_n$ is monotone increasing.

Question

Is constant sequence a monotonic sequence?

Monotone sequence: Convergence vs bounded

- Monotone sequence guarantee the next term is always larger/smaller than the previous.
- In this case, can we determine the convergence of a monotone sequence?
- We know that a sequence converges implies that it is bounded (Theorem 5.4). Now, how about a monotone sequence?

Theorem 7.3: Monotone Convergence Theorem (MCT)

Every bounded monotone sequence is convergent.

Remark:

(a) To be precise, every increasing sequence that is bounded above is convergent,
and every decreasing sequence that is bounded below is convergent,

(b) Must have bounded monotone sequence to converge.

(c) A constant sequence is a bounded monotone sequence, so it is convergent.

(d) Not every bounded sequence is convergent, but bounded monotone sequence is convergent.

Video is available!

How to use MCT?

- Step 1: Check if a sequence is monotone increasing(decreasing).

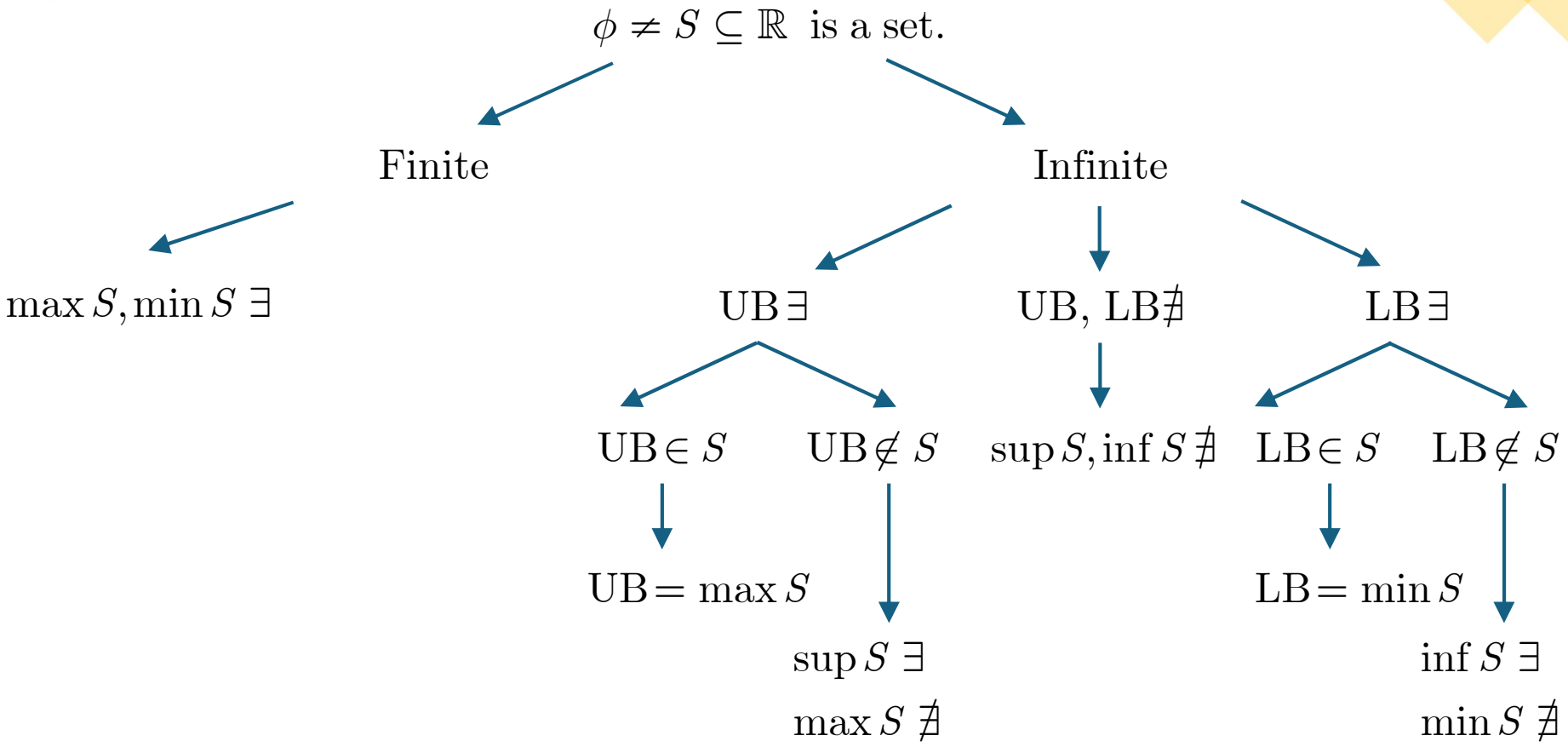


- Step 2: Check if the sequence is bounded above(below).



- Conclusion: The sequence is convergent.

Love affairs between UB – max – sup and LB – min – inf



Note that if $\max S = \sup S$ if maximum exists. Similarly for minimum.

A special theorem from sequences that links to sup/inf of sets

Supremum of a set vs limit of a sequence

- In Chapter 1, we studied supremum in a set S.
- If sequence (a_n) is constructed from taking elements of S, we ask: Is there any relation for a sup S and $\lim(a_n)$?
- The next theorem illustrate the relation.
- Note: In the process of constructing (a_n) , we are not required to take all (but we can if we want) elements from S. We can take some elements, but you may take more than once.

Theorem 7.5. (Supremum)

Let $\phi \neq S \subset \mathbb{R}$ be bounded above and let $\alpha \in \mathbb{R}$ be an upper bound for S. Then $\alpha = \sup S$ if and only if there exists a sequence (a_n) with $a_n \in S \; \forall n \in \mathbb{N}$ such that $\lim_{n\to\infty} a_n = \alpha$.

Theorem 7.5. (Infimum)

Let $\phi \neq S \subset \mathbb{R}$ be bounded below and let $\beta \in \mathbb{R}$ be a lower bound for S. Then $\beta = \inf S$ if and only if there exists a sequence (b_n) with $b_n \in S \; \forall n \in \mathbb{N}$ such that $\lim_{n\to\infty} b_n = \beta$.

Videos are available!

How to use Theorem 7.5?

- Step 1: Construct a convergent sequence and find $\lim(a_n) = a$.



- Step 2: Check if a is an upper bound (lower bound) of set S.



- Conclusion: If a is an upper bound, then $a = \sup S$.
  (If a is a lower bound, then $a = \inf S$.)

Note: You must know what is the relationship of S and a_n.

Example 7.6

Let $A = \left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\} \subset \mathbb{R}$.

Note that sup A exists because

- $0 \in A \Rightarrow A \neq \phi$
- $A \subset \mathbb{R}$ (given)
- $1 > 1 - \frac{1}{n} \forall n \in \mathbb{N} \Rightarrow A$ is bounded above

Define sequence $\left(a_n\right) = \left(1 - \frac{1}{n}\right)$.

Then $\lim\left(a_n\right) = 1$.

Note that 1 is an upper bound.

So by Theorem 7.5, sup $A = 1$.

Seriously? So easy to find the sup by using this Theorem?

Let’s apply the Theorem on some questions in Worksheet 1

Q3a) Worksheet 1

(a) Let $A = \left\{ \frac{m}{n} : m, n \in \mathbb{N}, m < n \right\}$.

Take $m = 1$ and construct $\left(a_n\right) = \left(\frac{1}{n+1}\right)$.

Verify that this sequence is decreasing and $0 < a_n < 1 \; \forall n \in \mathbb{N}$ (bounded).

Hence, by MCT, $\left(a_n\right)$ converges;

in fact, $\lim\left(a_n\right) = 0$.

Now note that 0 is a lower bound for A since $0 < m < n \Rightarrow 0 < \frac{m}{n} < 1$.

Hence, by Theorem 7.5 (inf), $\inf A = 0$.

Take $n = m + 1$ and construct $\left(\tilde{a}_m\right) = \left(\frac{m}{m+1}\right) = \left(1 - \frac{1}{m+1}\right)$.

Verify that this sequence is increasing and $0 < \tilde{a}_m < 1 \; \forall m \in \mathbb{N}$ (bounded).

Hence, by MCT, $\left(\tilde{a}_n\right)$ converges;

in fact, $\lim\left(\tilde{a}_m\right) = 1$.

Now note that 1 is an upper bound for $A = \left\{\frac{m}{n} : m, n \in \mathbb{N}, m < n\right\}$ since

$$0 < m < n \Rightarrow 0 < \frac{m}{n} < 1.$$

Hence, by Theorem 7.5 (supremum), $\sup A = 1$.

Q3b) Worksheet 1 (when things did not go well!)

(b) Let $B = \left\{ \frac{(-1)^m}{n} : m, n \in \mathbb{N} \right\} = \left\{ \pm \frac{1}{n} : n \in \mathbb{N} \right\} = \left\{ \pm 1, \pm \frac{1}{2}, \pm \frac{1}{3}, \ldots \right\}$.

If we construct $\left(b_n\right) = \left(\frac{1}{n}\right)$ and $\left(\tilde{b}_n\right) = \left(-\frac{1}{n}\right)$.

The sequences are bounded and monotone.

So we can apply MCT to show these sequences converge.

Note that $\lim\left(b_n\right) = 0 = \lim\left(\tilde{b}_n\right)$.

But 0 is neither an upper bound nor a lower bound of B.

So, Theorem 7.5 is failed to apply when we consider these sequences.

Let $n = 1$ and m be even, then we have a sequence $(1, 1, 1, \ldots)$.

Let $n = 1$ and m be odd, then we have a sequence $(-1, -1, -1, \ldots)$.

Obviously these two sequences converge to 1 and -1, respectively.

Note that $-1 \leq \frac{(-1)^m}{n} \leq 1 \; \forall m, n \in \mathbb{N}$.

So -1 is a lower and 1 is an upper bound.

Hence by Theorem 7.5 (sup) (inf), we have $\sup B = 1$ and $\inf B = -1$.

Q3c) Worksheet 1

(c) Let $C = \left\{ \frac{n}{3n+1} : n \in \mathbb{N} \right\}$. Construct $\left(c_n \right) = \left(\frac{n}{3n+1} \right)$.

Now note that $\left(c_n \right)$ is monotonic increasing ($\because c_{n+1} - c_n = \frac{1}{(3n+1)(3n+4)} > 0$)

$\left(c_n \right)$ is bounded ($\because 0 < c_n = \frac{1}{3}\left(1 - \frac{1}{3n+1}\right) < \frac{1}{3} \ \forall n \in \mathbb{N}$).

By MCT, $\left(c_n \right)$ is convergent; in fact, $\lim\left(c_n \right) = \frac{1}{3}$.

Now $\frac{1}{3}$ is an upper bound for C.

Hence, by Theorem 7.5 (sup), $\sup C = \frac{1}{3}$.

Recall if a lower bound is in the set, then that lower bound is the min

Since $\left(c_n\right)$ is monotonic increasing,

It follows that $\frac{1}{4} = c_1 \leq c_n \forall n \in \mathbb{N}$.

So $\frac{1}{4}$ is a lower bound but $\frac{1}{4} \in C$,

which means $\min C = \frac{1}{4} = \inf C$.

Q3d) Worksheet 1

(d) Let $D = \left\{ \frac{m}{m+n} : m, n \in \mathbb{N} \right\}$.

Take $m = 1$ and construct $\left(d_n\right) = \left(\frac{1}{1+n}\right)$.

Verify that $\left(d_n\right)$ is decreasing and $0 < d_n < 1 \; \forall n \in \mathbb{N}$ (bounded).

So, by MCT, $\left(d_n\right)$ converges.

In fact, $\lim\left(d_n\right) = 0$.

Now $0 < d \; \forall d \in D$, which mean 0 is a lower bound.

So, by Theorem 7.5 (inf), inf $D = 0$.

Note that we apply Theorem 7.5 and MCT

Take $n = 1$ and construct $\left(\tilde{d}_m\right) = \left(\frac{m}{m+1}\right)$.

Verify that $\left(\tilde{d}_n\right)$ is increasing and $0 < d_n < 1 \; \forall n \in \mathbb{N}$ (bounded).

So, by MCT, $\left(\tilde{d}_n\right)$ converges.

In fact, $\lim\left(\tilde{d}_n\right) = 1$.

Now $d < 1 \; \forall d \in D$, which mean 1 is an upper bound.

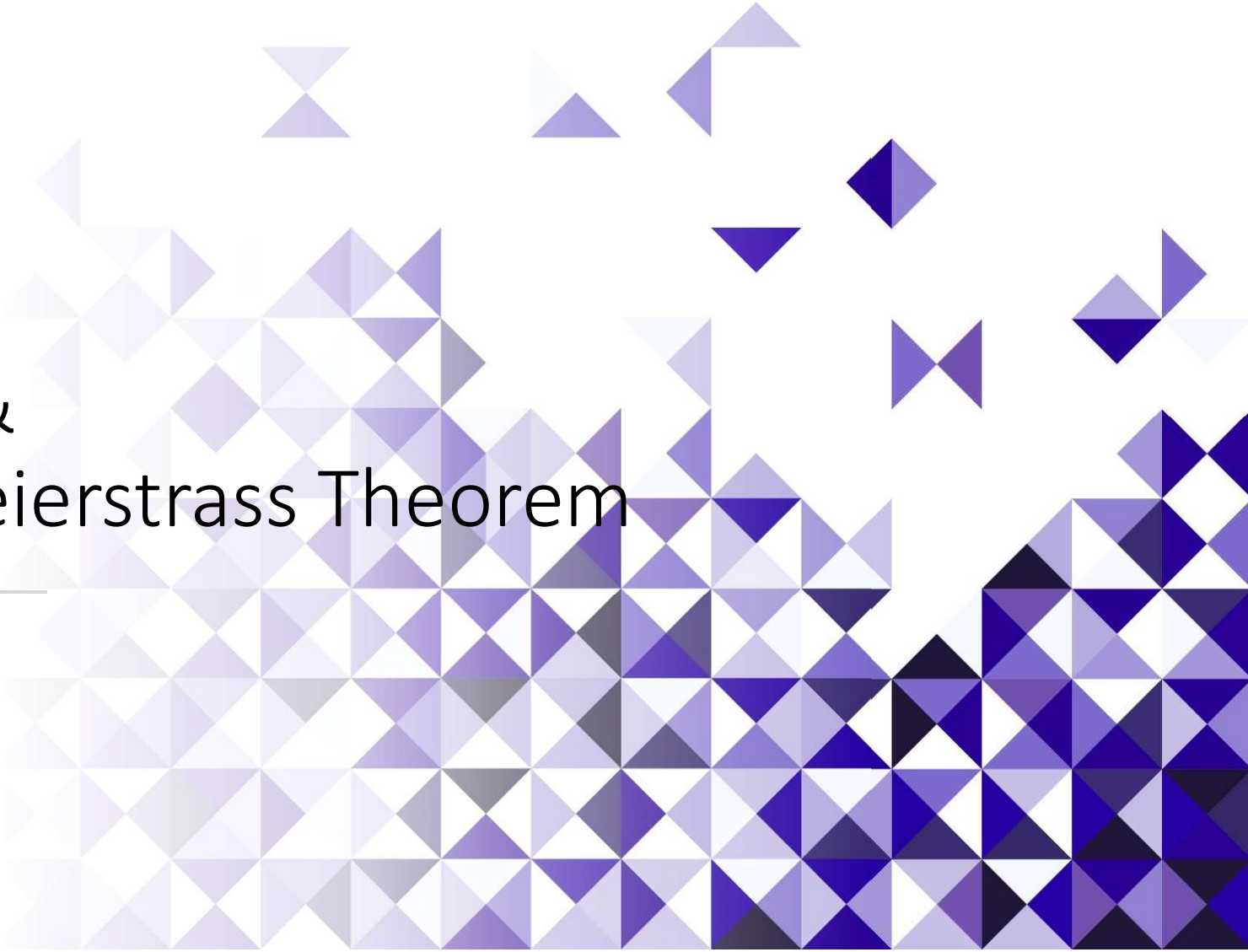
So, by Theorem 7.5 (sup), sup $D = 1$.

Example 7.4. Show that the sequence (b_n) given by $b_{n+1} = \frac{1}{4-b_n}$, $b_1 = 3$ is convergent and then find its limit.

Check out the video to see how we can prove that this sequence is convergent even if we do not find the general explicit formula for b_n !!!

n	b_n
1	3
2	1
3	0.333333
4	0.272727
5	0.268293
6	0.267974
7	0.267951
8	0.267949
9	0.267949
10	0.267949

Subsequences & The Bolzano-Weierstrass Theorem



Definition

A subsequence $\left(a_{n_k}\right)$ can be constructed if the following conditions are met:

1. $1 \leq n_1 < n_2 < \cdots < n_k$
2. $n_k \in \mathbb{N}\ \forall k \in \mathbb{N}$

We can also write $\left(a_{n_k}\right) = \left(\tilde{a}_k\right)$, where $\left(\tilde{a}_k\right)$ is a new sequence (subsequence of (a_n) with subscript $k = 1, 2, 3, \ldots$).

For example, we construct :

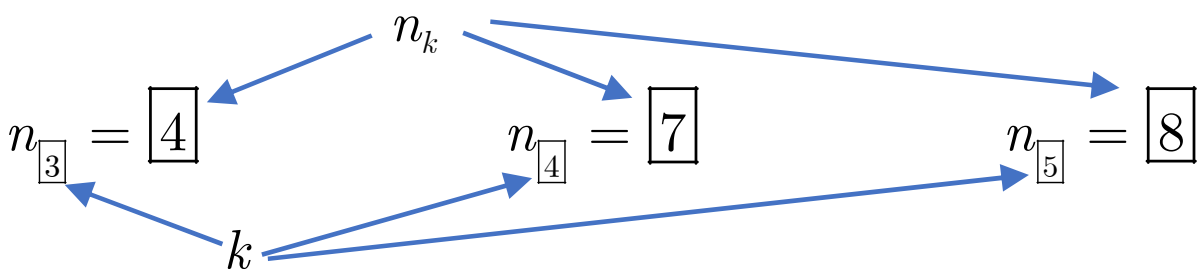
$$
\begin{array}{rccccccccccc}
(a_n): & a_1 & a_2 & a_3 & a_4 & a_5 & a_6 & a_7 & a_8 & \cdots & a_n \\
 & \| & \| & & \| & & & \| & \| & & \| \\
(a_{n_k}): & a_{n_1} & a_{n_2} & & a_{n_3} & & & a_{n_4} & a_{n_5} & \cdots & a_{n_k} \\
 & (n_1=1) & (n_2=2) & & (n_3=4) & & & (n_4=7) & (n_5=8) & & (n_k=n) \\
 & \| & \| & & \| & & & \| & \| & & \| \\
(a_{n_k}) = (\tilde{a}_k): & \tilde{a}_1 & \tilde{a}_2 & & \tilde{a}_3 & & & \tilde{a}_4 & \tilde{a}_5 & \cdots & \tilde{a}_k \\
 & (k=1) & (k=2) & & (k=3) & & & (k=4) & (k=5) & &
\end{array}
$$

Therefore, $\boxed{n_k \geq k}$, where $n_k = k$ if we select the $a_1, a_2, \ldots, a_k$ (k consecutive elements) of (a_n) for the subsequence (a_{n_k}).

Observation

In another words,

$n_1 = 1$ $n_2 = 2$



We see that $n_1 = 1$, $n_2 = 2$, $n_3 = 4 > 3$, $n_4 = 7 > 4$, $n_5 = 8 > 5$. In general,

$$\boxed{n_k \geq k}.$$

Example 8.2 (i) Let $(a_n) = ((-1)^n)$

(a_n): $a_1 = -1$ $\boxed{a_2 = 1}$ $a_3 = -1$ $\boxed{a_4 = 1}$ $\cdots$

(a_{n_k}): $a_{n_1} = 1$ $(n_1 = 2)$ $a_{n_2} = 1$ $(n_2 = 4)$ $\cdots$

$$(n_k) = (2, 4, 6, \ldots)$$

$$(a_{n_k}) = (1, 1, 1, \ldots)$$

We can form a pattern on selecting the sequence to form a subsequence, for example, take all the even terms!

Example 8.2 (ii) Let $(b_n) = (1/n)$

In this example, we take all the 10^n–th terms!

(ii) Let (b_n) be the sequence given by $b_n = 1/n$. Then the sequence given by

$$(n_k) = (10, 100, 1000, \dots)$$

gives the subsequence

$$(b_{n_k}) = \left(\frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \dots\right).$$

Main theorem and its corollary

Theorem 8.3. (a_n) converges iff every subsequence of (a_n) converges (to the same limit as (a_n) converges to).

$$\left(a_n\right) \text{ converges (to } a) \Leftrightarrow \underline{\text{all}} \left(a_{n_k}\right) \text{ converge (to } a)$$

Logically equivalent to

$$\left(a_n\right) \text{ diverges } \Leftrightarrow \text{there exists divergent } \left(a_{n_k}\right) \text{ or two } \left(a_{n_k}\right) \text{ converge differently.}$$

Corollary 8.4. If there are two subsequences of (a_n) converge to a different limit, then (a_n) is divergent.

Video is available!

How to apply Corollary 8.4?

- Step 1: Construct two subsequences such that their limits are distinct.



- Conclusion: The original sequence diverges.

Example 8.5. Show that $(a_n) = ((-1)^n)$ diverges.

Proof. Consider $(a_n) = ((-1)^n)$. Then we can form two subsequences

$$\left(a_{n_k}\right) = (1,1,1,\ldots) \qquad \text{where } (n_k) = (2,4,6,\ldots)$$
$$\left(a_{\tilde{n}_k}\right) = (-1,-1,-1,\ldots) \qquad \text{where } (\tilde{n}_k) = (1,3,5,\ldots)$$

But note that

$$\left(a_{n_k}\right) = (1,1,1,\ldots) \to 1 \qquad \text{when } n \to \infty$$
$$\left(a_{\tilde{n}_k}\right) = (-1,-1,-1,\ldots) \to -1 \qquad \text{when } n \to \infty$$

So $\lim\left(a_{n_k}\right) \neq \lim\left(a_{\tilde{n}_k}\right) \Rightarrow \left(a_n\right)$ diverges.

Extra example (a)

Use Corollary 8.4 to show that the sequences below are divergent.

(a) $\left(1+(-1)^n\right)$ (b) $\left(\cos n\pi\right)$ (c) (1, 1, 1, 2, 1, 3, 1, 4, 1, 5, 1, 6, ...)

Solution. (a) Choose the following two subsequences

$$\left(a_{n_k}\right) = (0,\ 0,\ 0,\ \ldots\ ) \text{ where } \left(n_k\right) = \left(1,3,5,\ldots\right)$$

$$\text{and } \left(a_{\tilde{n}_k}\right) = (2,\ 2,\ 2,\ \ldots) \text{ where } \left(\tilde{n}_k\right) = \left(2,4,6,\ldots\right).$$

One subsequence converges to 0 and the other one to 1. Hence, by Corollary 8.4, $\left(1+(-1)^n\right)$ diverges.

Extra example (b)

(b) $\left(\cos n\pi\right)$

Choose the following two subsequences

$$\left(a_{n_k}\right) = (-1, -1, -1, \ldots) \text{ where } \left(n_k\right) = \left(1, 3, 5, \ldots\right)$$

$$\text{and } \left(a_{\tilde{n}_k}\right) = (1, 1, 1, \ldots) \text{ where } \left(\tilde{n}_k\right) = \left(2, 4, 6, \ldots\right).$$

One subsequence converges to –1 and the other one to 1. Hence, by Corollary 8.4, $\left(\cos n\pi\right)$ diverges.

Extra example (c)

(c) (1, 1, 1, 2, 1, 3, 1, 4, 1, 5, 1, 6, ...)

Choose the subsequence (1, 2, 3, 4, 5, 6, ...) where $\left(\tilde{n}_k\right) = \left(2, 4, 6, ...\right)$.

Then the subsequence diverges (to ∞) and hence (1, 1, 1, 2, 1, 3, 1, 4, 1, 5, 1, 6, ...) is divergent.

Theorem 8.6. Every sequence has a monotone subsequence.

Here it means no matter what sequence you choose (whether bounded or not, convergent or not), you will always get a monotone subsequence.

This theorem is used to prove the next important theorem in Analysis.

Video is available!

Examples

Remark: If $\left(a_n\right)$ is a monotonic sequence then choose the subsequence $\left(a_{n_k}\right) = \left(a_n\right)$.

1. $a_n = \frac{3n+1}{n+2}$ is increasing for all natural numbers n.
2. $a_n = \frac{3n^2+2n-10}{n^2-3n-3}$ is decreasing when $n > 3$. Choose $\left(n_k\right) = \left(4,5,6,...\right)$.
3. $a_n = n^2$ is increasing for all natural numbers n.
4. $a_n = (-1)^n$. Choose $\left(n_k\right) = \left(2,4,6,...\right)$.
5. $a_n = c =$ constant, which is both increasing and decreasing.
6. $a_n = (-1)^n n$. Choose $\left(n_k\right) = \left(2,4,6,...\right)$.

Theorem 8.7
Bolzano-Weierstrass Theorem

I cannot tell how rich you are, but I can tell at least one of your child is rich.

Theorem 8.7: Bolzano-Weierstrass Theorem (BWT)

Every bounded sequence has a convergent subsequence.

Proof idea.

Fact. All subsequence of a bounded sequence is bounded.

Theorem 8.6. Every sequence has a monotone subsequence.

Theorem 7.3 (MCT). Every bounded monotone sequence is convergent.

Video is available!

What is so special about this BWT?

Check out the last question in Worksheet 3.

Q17 in Worksheet 3.

Prove that every nonempty subset of the real numbers that is bounded above has a supremum directly from the Bolzano-Weierstrass theorem.

Is this enough to make you say “WOOOWWWW!!” ?

Example: (a) $a_n = (-1)^n$ is bounded so it has a convergent subsequence,

$(1, 1, 1, \ldots)$ or $(-1, -1, -1, \ldots)$

b) When a sequence is unbounded, we cannot apply BWT. We can actually find convergent and/or divergent subsequence.

For example, $a_n = \begin{cases} \frac{1}{n} & n \text{ is odd} \\ n & n \text{ is even} \end{cases}$. In this case,

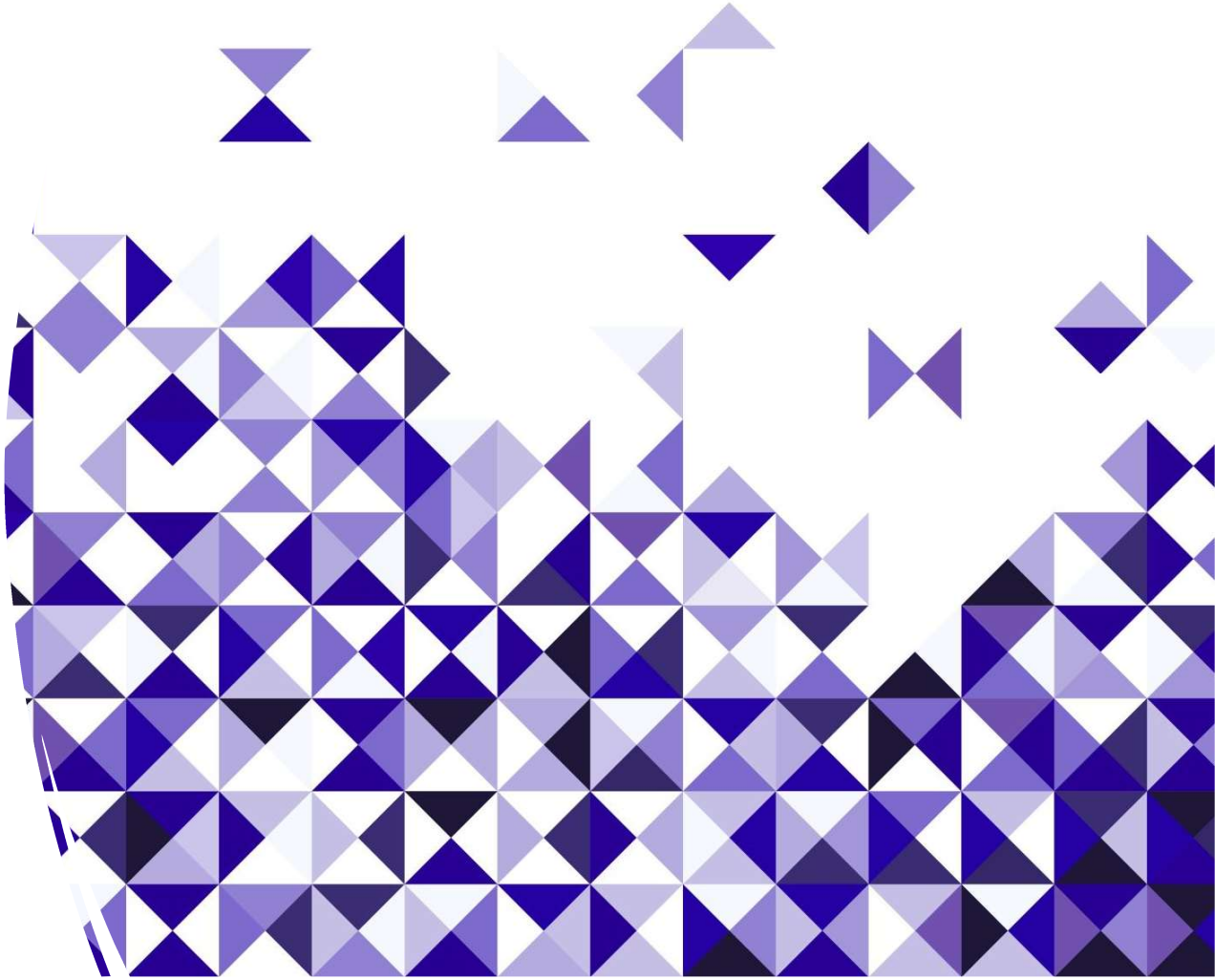
when $\left(n_k\right) = \left(2, 4, 6, \ldots\right)$, we can form a divergent subsequence (2, 4, 6, …),

and when $\left(\tilde{n}_k\right) = \left(1, 3, 5, \ldots\right)$, a convergent subsequence $\left(1, \frac{1}{3}, \frac{1}{5}, \ldots\right)$.

A sequence is		the sequence is	True/False
convergent	$\Rightarrow$	bounded	T
bounded	$\Rightarrow$	convergent	F
divergent	$\Rightarrow$	unbounded	F
unbounded	$\Rightarrow$	divergent	T
monotone	$\Rightarrow$	convergent	F
bounded & monotone	$\Rightarrow$	convergent	T
Cauchy	$\Rightarrow$	bounded	T
bounded	$\Rightarrow$	Cauchy	F
Cauchy	$\Rightarrow$	convergent	T
convergent	$\Rightarrow$	Cauchy	T

A sequence is		$\exists$ / $\forall$ subsequence(s) is/are	True/False
convergent	$\Rightarrow$	convergent (to the same limit)	
convergent	$\Rightarrow$	convergent (to different limit)	
bounded	$\Rightarrow$	bounded	
unbounded	$\Rightarrow$	unbounded	
arbitrary	$\Rightarrow$	monotone	
arbitrary	$\Rightarrow$	monotone & bounded	
bounded	$\Rightarrow$	convergent	
unbounded	$\Rightarrow$	divergent	

The Cauchy sequence



Motivation

- Another approach to look at the convergence of a sequence:
- We investigate the relation of two elements of the sequence after n becomes arbitrarily large.
- In other words, we look at $|a_m - a_n|$ after a point of time, say N, meaning for m, $n > N$, what would $|a_m - a_n|$ be?
- What do you think?

Definition 9.1. Let (a_n) be a sequence. Then (a_n) is said to be a Cauchy sequence (or (a_n) is Cauchy in short) if $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that

$$\left|a_m - a_n\right| < \varepsilon$$

whenever $m, n > N$.

Remarks:

- A **Cauchy sequence** is a sequence whose elements become arbitrarily close to each other as the sequence progresses.
- Cauchy sequences are named after Augustin-Louis Cauchy; they may occasionally be known as **fundamental sequences**.

Why not $|a_{n+1} - a_n| < \varepsilon$?

$\left(\sqrt{n}\right)$ is a divergent sequence even though it seems like a_n and a_{n+1} are getting closer and closer when $n \to \infty$.

$$\left|a_{n+1} - a_n\right| = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{2\sqrt{n}} < \varepsilon\,.$$

This is a classical counter example that motivates the definition of Cauchy sequence.

Cauchy sequence is my favourite sequence!!

- If one can shows that a sequence is Cauchy, then we have two good news to tell!!
- If a sequence is Cauchy, then the sequence is BOUNDED and CONVERGENT!
- The next two theorems show us this fantastic results!

Theorem 9.2. All Cauchy sequences are bounded.

Theorem 9.3. (a_n) is Cauchy iff (a_n) is convergent.

Obviously if we have Theorem 9.3 then Theorem 9.2 is true (because convergent sequences are bounded).
However, we prove Theorem 9.3 by using Theorem 9.2 (and Theorem 8.7), so Theorem 9.2 is essential.

Videos are available!

Theorem 9.3 is also known as the Cauchy Criterion (for sequence)

A sequence converges if and only if it is a Cauchy sequence.

Example. Use the definition to prove the following sequences are Cauchy. Then find their limits.

(a) $\left(a_n\right) = \left(5 - \frac{3}{n}\right)$

(b) $\left(b_n\right) = \left(\frac{\sin n}{n}\right)$

Proof. Recall the triangle inequality: $\left|x-y\right| \leq \left|x\right| + \left|-y\right| = \left|x\right| + \left|y\right|$.

(a) By Archimedean property: $\forall\, \frac{\varepsilon}{6} > 0\ \exists N \in \mathbb{N}$ э $\frac{\varepsilon}{6} > \frac{1}{N} \Rightarrow \varepsilon > \frac{6}{N}$.

Now for all $n, m > N$, it implies that

(a_n) is Cauchy
$\Rightarrow (a_n)$ converges

$$\left|a_m - a_n\right| = \left|\tfrac{3}{n} - \tfrac{3}{m}\right| = \tfrac{3}{mn}\left|m-n\right|$$
$$\leq \tfrac{3}{mn}\left(\left|m\right| + \left|n\right|\right) = \tfrac{3}{n} + \tfrac{3}{m} < \tfrac{3}{N} + \tfrac{3}{N} < \varepsilon.$$

Furthermore, $\lim\left(a_n\right) = \lim\left(5 - \tfrac{3}{n}\right) = 5$ (by Theorem 6.1).

(b) By Archimedean property: $\forall\, \frac{\varepsilon}{2} > 0\ \exists N \in \mathbb{N}\ \ni \frac{\varepsilon}{2} > \frac{1}{N} \Rightarrow \varepsilon > \frac{2}{N}$.

Now for all $n, m > N$, it implies that

$$\left|b_m - b_n\right| = \left|\frac{\sin m}{m} - \frac{\sin n}{n}\right| \leq \left|\frac{\sin m}{m}\right| + \left|\frac{\sin n}{n}\right| \leq \frac{1}{m} + \frac{1}{n} < \frac{1}{N} + \frac{1}{N} < \varepsilon.$$

Furthermore, $\lim\left(b_n\right) = \lim\left(\frac{\sin n}{n}\right) = 0$ (Example 6.5(i)).

Example 9.4. Can we use MCT to show the convergence of the sequence $a_{n+1} = 1 + \frac{1}{a_n}$ $\forall n \in \mathbb{N}$ with $a_1 = 1$?

Answer is no. Note that

$$a_2 = 1 + \frac{1}{a_1} = 2 > a_1 \geq 1$$
$$a_3 = 1 + \frac{1}{a_2} = \frac{3}{2} < a_2$$
$$a_4 = 1 + \frac{1}{a_3} = \frac{5}{3} < a_3$$

Generated by Excel:

n	1	2	3	4	5	6	7	8	9	10
a_n	1	2	1.5	1.666667	1.6	1.625	1.615385	1.619048	1.617647	1.618182

n	11	12	13	14	15	16	17	18
a_n	1.617978	1.618056	1.618026	1.618037	1.618033	1.618034	1.618034	1.618034

Example 9.4: Cauchy criterion for convergence

Let (a_n) be a sequence given by

$$a_{n+1} = 1 + \frac{1}{a_n}$$

and $a_1 = 1$. Then (a_n) is Cauchy and hence convergent.

Video is available!

Summary

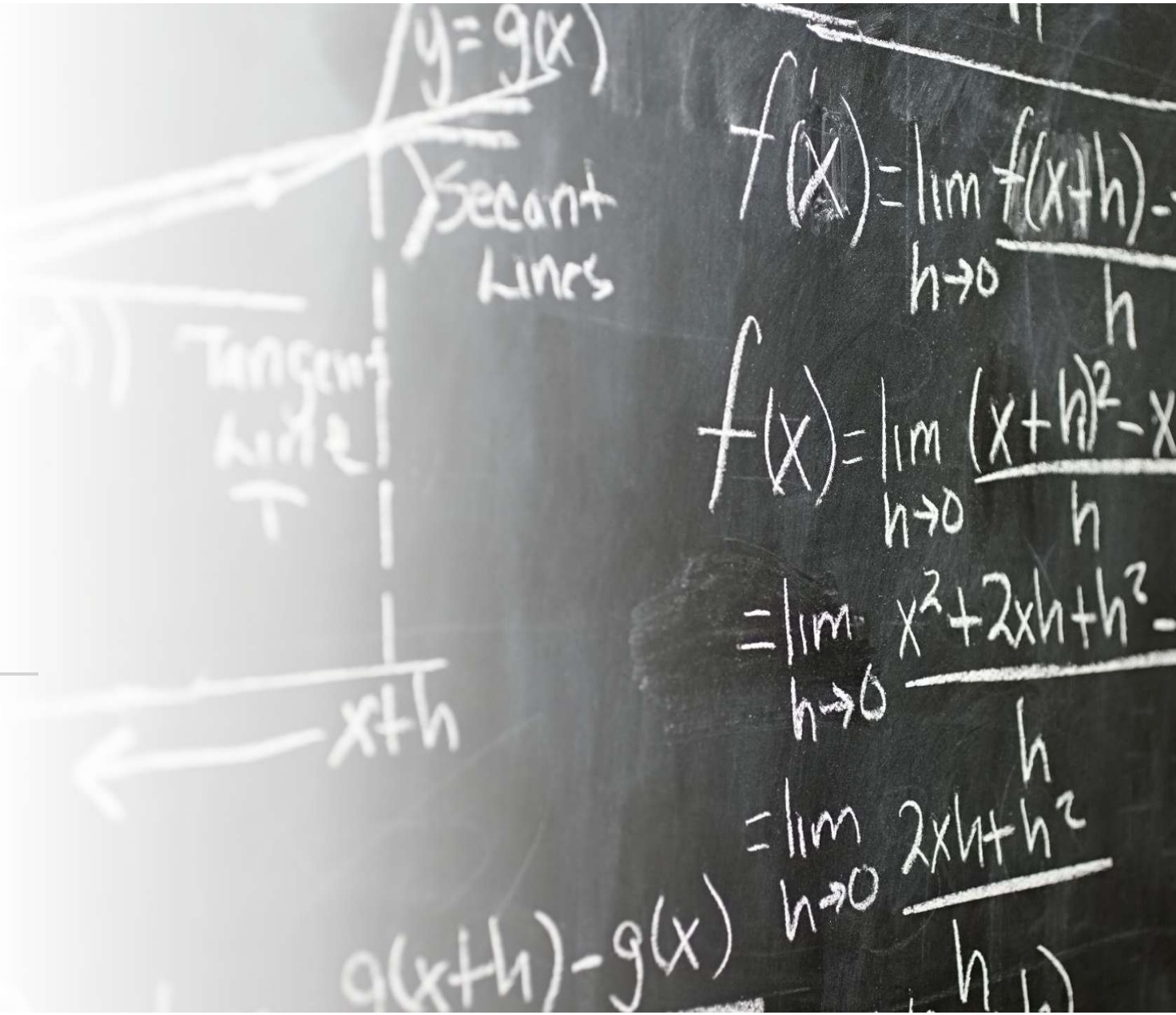
- Two main theorems here:

Let (a_n) be a sequence.

$$(a_n) \text{ is Cauchy} \Rightarrow (a_n) \text{ is bounded.}$$

$$(a_n) \text{ is Cauchy} \Leftrightarrow (a_n) \text{ is convergent.}$$

Special sequence that involving sums



Partial Sum

Let $\left(a_k\right) = \left(a_1, a_2, a_3, \ldots\right)$ be a sequence.

Suppose $\left(s_k\right) := \left(a_1, a_1 + a_2, a_1 + a_2 + a_3, \ldots\right)$.

Note that $s_k = \sum_{i=1}^{k} a_i$ and we call this as the partial sum.

Series

If the sequence $\left(s_k\right)$ converges, then

$$\lim\left(s_k\right) = \lim\left(\sum_{i=1}^{k} a_i\right) = \lim_{k\to\infty}\left(\sum_{i=1}^{k} a_i\right) = \sum_{i=1}^{\infty} a_i \ ,$$

and we call this as an infinite sum, or simply a (convergent) series.

If the sequence $\left(s_k\right)$ diverges, then the series $\sum_{i=1}^{\infty} a_i$ also diverges.

Example

Let $\left(a_r\right) = \left(1, 2, 3, \ldots\right)$. Then

$$s_k = \sum_{r=1}^{k} a_r = \sum_{r=1}^{k} r = \tfrac{k}{2}(k+1) \text{ is the partial sum.}$$

Since $\lim\left(s_k\right) = \sum_{r=1}^{\infty} a_r = \sum_{r=1}^{\infty} r = \lim\left(\tfrac{k}{2}(k+1)\right) = \infty$,

it follows that the series $\sum_{r=1}^{\infty} a_r$ diverges.

Example

Let $\left(a_r\right) = \left(1^2, 2^2, 3^2, \ldots\right)$. Then

$$s_k = \sum_{r=1}^{k} r^2 = \tfrac{k}{6}(k+1)(2k+1)$$

$$\Rightarrow \lim\left(s_k\right) = \sum_{r=1}^{\infty} a_r = \sum_{r=1}^{\infty} r^2 = \lim\left(\tfrac{k}{6}(k+1)(2k+1)\right) = \infty.$$

Hence, the series $\sum_{r=1}^{\infty} a_r$ diverges.

Example

Let $\left(a_r\right) = \left(\frac{1}{r^2+r}\right)$. Then

$$s_k = \sum_{r=1}^{k} \frac{1}{r^2+r} = 1 - \frac{1}{k+1}$$

$$\Rightarrow \lim\left(s_k\right) = \sum_{r=1}^{\infty} a_r = \sum_{r=1}^{\infty} \frac{1}{r^2+r} = \lim\left(1 - \frac{1}{k+1}\right) = 1$$

Hence, the series converges.

Example

Let $\left(a_r\right) = \left(2, 2^2, 2^3, \ldots\right)$. Then

$$\left(s_k\right) = \sum_{r=1}^{k} 2^r = 2(2^k - 1)$$

$$\Rightarrow \lim\left(s_k\right) = \sum_{r=1}^{\infty} 2^r = \lim\left(2(2^k - 1)\right) = \infty$$

Hence, the series diverges.

Exercise

Show that $\sum_{r=1}^{\infty}\left(\frac{1}{2}\right)^r$ is a convergent series.

Summary



Let (a_n) be a sequence.

Theorem 5.4. If (a_n) is convergent, then (a_n) is bounded.

Theorem 7.3. (i) If (a_n) is increasing and bounded above then (a_n) is convergent.

(ii) If (a_n) is decreasing and bounded below then (a_n) is convergent.

Theorem 8.3. (a_n) converges to L iff every subsequence of (a_n) converge to L.

Corollary 8.4. If there are two subsequences of (a_n) converge to different limit, then (a_n) is divergent.

Theorem 8.6. Every sequence has a monotone subsequence.

Theorem 8.7. Every bounded sequence has a convergent subsequence.

Theorem 9.2. All Cauchy sequences are bounded.

Theorem 9.3. (a_n) is Cauchy iff (a_n) is convergent.

To determine whether a given (a_n) is convergent?

Check 1: Is the sequence unbounded? If yes, then it is divergent.

Check 2: Is the sequence monotone increasing/decreasing that bounded above/below. If yes, then it is convergent.

Check 3: Can we find two subsequences of (a_n) such that they converge to different limit? If yes, then (a_n) is divergent.

Check 4: Is (a_n) a Cauchy sequence? If yes, then (a_n) is convergent. If no, then (a_n) is divergent.

Rotate your head or the tablet?

